1 Simplex and Matrices

We will begin with a review of matrix multiplication. A matrix is simply an array of numbers. If a given array has m rows and n columns, then it is called an $m \times n$ (or m-by-n) matrix. As examples, the array

$$\mathbf{A}_1 = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 2 \end{bmatrix}$$

is a 2×3 matrix; and the array

$$\mathbf{A}_2 = \begin{bmatrix} 2 & 0 & 4 \\ -1 & 3 & 2 \\ 5 & 1 & -2 \end{bmatrix}$$

is a 3×3 matrix.

A number can be viewed as a 1×1 matrix. We are, of course, familiar with the multiplication of numbers. The multiplication between two matrices is an operation that is an extension of the multiplication operation between two numbers. This extension, however, will not be applicable to all pairs of matrices. More specifically, let $\mathbf{A}$ and $\mathbf{B}$ be two matrices; then, the product of these two matrices, denoted by $\mathbf{A} \cdot \mathbf{B}$, is defined only when the number of columns in $\mathbf{A}$ is the same as the number of rows in $\mathbf{B}$. Since the matrices $\mathbf{A}_1$ and $\mathbf{A}_2$ above satisfy this condition, we will first construct the product of these two matrices as a specific numerically example.

Denote by $\mathbf{A}_3$ the product of $\mathbf{A}_1$ and $\mathbf{A}_2$. The number of rows in $\mathbf{A}_3$ will be identical to that of $\mathbf{A}_1$; and the number of columns in $\mathbf{A}_3$ will be identical to that of $\mathbf{A}_2$. That is, $\mathbf{A}_3$ will have 2 rows and 3 columns. We will construct the rows in $\mathbf{A}_3$ one by one. The first row in $\mathbf{A}_3$ will be determined by entries in the first row in $\mathbf{A}_1$ and all of the rows in $\mathbf{A}_2$, as follows. Denote the entries in the first row of $\mathbf{A}_1$ by a_{11}, a_{12}, and a_{13}; and denote the rows in $\mathbf{A}_2$ by R_1, R_2, and R_3. Then, the first row in $\mathbf{A}_3$ is defined to be the outcome of the row operations $a_{11} \cdot R_1 + a_{12} \cdot R_2 + a_{13} \cdot R_3$. Explicitly, these operations yield

$$1 \cdot \begin{bmatrix} 2 & 0 & 4 \end{bmatrix} + (-1) \cdot \begin{bmatrix} -1 & 3 & 2 \end{bmatrix} + 0 \cdot \begin{bmatrix} 5 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 3 & -3 & 2 \end{bmatrix}.$$

In a similar way, the second row of $\mathbf{A}_3$ will be determined by entries in the second row in $\mathbf{A}_1$ and all of the rows in $\mathbf{A}_2$ as follows.

$$(-2) \cdot \begin{bmatrix} 2 & 0 & 4 \end{bmatrix} + 3 \cdot \begin{bmatrix} -1 & 3 & 2 \end{bmatrix} + 2 \cdot \begin{bmatrix} 5 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 3 & 11 & -6 \end{bmatrix}.$$

Hence, the product of $\mathbf{A}_1$ and $\mathbf{A}_2$ is given by

$$\mathbf{A}_3 = \begin{bmatrix} 3 & -3 & 2 \\ 3 & 11 & -6 \end{bmatrix}.$$

Although the above numerical illustration of the definition of matrix multiplication is actually sufficient for our modest purposes here, we will quickly summarize these calculations in a formal definition, for the sake of completeness. Let $\mathbf{C}$ denote the product of two matrices $\mathbf{A}$ and $\mathbf{B}$. Suppose $\mathbf{A}$ is $m \cdot k$ and $\mathbf{B}$ is $k \cdot n$; and let a_{ij} and b_{ij} denote the respective entries at the intersection of the ith row and the jth column in $\mathbf{A}$ and $\mathbf{B}$. Then, $\mathbf{C}$ is defined as the $m \cdot n$ matrix whose entry at the intersection of its ith row and its jth column is specified by

$$c_{ij} \equiv \sum_{l=1}^{k} a_{il} b_{lj} .$$

What is the connection between matrix multiplication and the Simplex algorithm? Observe that in the calculation of the product of $\mathbf{A}_1$ and $\mathbf{A}_2$, entries in the first row of $\mathbf{A}_1$, namely 1, -1, and 0, serve as individual multipliers to R_1, R_2, and R_3; and entries in the second row of $\mathbf{A}_1$, namely -2, 3, and 2, serve as a second set of multipliers to R_1, R_2, and R_3. With this perspective, we see that the language of matrix multiplication offers a very compact description of sets of row operations.

Indeed, our next step is to show that each pivot in the Simplex algorithm is equivalent to pre-multiplying a given tableau by an appropriately-chosen matrix. To understand what this statement means, we will revisit the linear program below, and use it as a concrete example.

$$
\begin{array}{lrrrrrrrcll}
\text{Maximize} & z \\
\text{Subject to:} \\
& z & -4x_1 & -3x_2 & & & & & = & 0 & (0)\\
& & 2x_1 & +3x_2 & +s_1 & & & & = & 6 & (1)\\
& & -3x_1 & +2x_2 & & +s_2 & & & = & 3 & (2)\\
& & & 2x_2 & & & +s_3 & & = & 5 & (3)\\
& & 2x_1 & +x_2 & & & & +s_4 & = & 4 & (4)
\end{array}
$$

$$x_1,\, x_2,\, s_1,\, s_2,\, s_3,\, s_4 \geq 0\,.$$

Ignoring the variable names, the initial tableau for this problem is:

1	-4	-3	0	0	0	0	0
0	2	3	1	0	0	0	6
0	-3	2	0	1	0	0	3
0	0	2	0	0	1	0	5
0	2	1	0	0	0	1	4

We will view this tableau as a matrix with 5 rows and 8 columns; and we will refer to it as $\mathbf{T}_I$. Recall that the pivot element in the ensuing pivot is the entry "2", located at the intersection of the last row and the second column; and that the specific row operations performed in this pivot are: $2 \cdot R_4 + R_0$, $(-1) \cdot R_4 + R_1$, $(3/2) \cdot R_4 + R_2$, $0 \cdot R_4 + R_3$, and $(1/2) \cdot R_4$. Consider the first set of operations, namely $2 \cdot R_4 + R_0$. Observe that this "recipe" can be rewritten as $1 \cdot R_0 + 0 \cdot R_1 + 0 \cdot R_2 + 0 \cdot R_3 + 2 \cdot R_4$. In other words, we can explicitly indicate the nonparticipation of R_1, R_2, and R_3 in these operations by introducing three new multipliers that are equal to 0. This augmented recipe provides a more-complete description of the operations, in that the level of participation of *every* row in $\mathbf{T}_I$ is explicitly indicated via an associated multiplier. Similarly, the operations $(-1) \cdot R_4 + R_1$ can be rewritten as $0 \cdot R_0 + 1 \cdot R_1 + 0 \cdot R_2 + 0 \cdot R_3 + (-1) \cdot R_4$; the operations $(3/2) \cdot R_4 + R_2$, as $0 \cdot R_0 + 0 \cdot R_1 + 1 \cdot R_2 + 0 \cdot R_3 + (3/2) \cdot R_4$; the operations $0 \cdot R_4 + R_3$, as $0 \cdot R_0 + 0 \cdot R_1 + 0 \cdot R_2 + 1 \cdot R_3 + 0 \cdot R_4$; and the operation $(1/2) \cdot R_4$, as $0 \cdot R_0 + 0 \cdot R_1 + 0 \cdot R_2 + 0 \cdot R_3 + (1/2) \cdot R_4$. Thus, to produce the new R_0, we perform the matrix multiplication

$$
\begin{bmatrix} 1 & 0 & 0 & 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & -4 & -3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 3 & 1 & 0 & 0 & 0 & 6 \\ 0 & -3 & 2 & 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 2 & 0 & 0 & 1 & 0 & 5 \\ 0 & 2 & 1 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 & 0 & 0 & 0 & 2 & 8 \end{bmatrix};
$$

and similarly, to produce the remaining new rows, we perform the matrix multiplications:

$$
\begin{bmatrix} 0 & 1 & 0 & 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 & -4 & -3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 3 & 1 & 0 & 0 & 0 & 6 \\ 0 & -3 & 2 & 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 2 & 0 & 0 & 1 & 0 & 5 \\ 0 & 2 & 1 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 2 & 1 & 0 & 0 & -1 & 2 \end{bmatrix},
$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 3/2 \end{bmatrix} \cdot \begin{bmatrix} 1 & -4 & -3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 3 & 1 & 0 & 0 & 0 & 6 \\ 0 & -3 & 2 & 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 2 & 0 & 0 & 1 & 0 & 5 \\ 0 & 2 & 1 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 7/2 & 0 & 1 & 0 & 3/2 & 9 \end{bmatrix},$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & -4 & -3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 3 & 1 & 0 & 0 & 0 & 6 \\ 0 & -3 & 2 & 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 2 & 0 & 0 & 1 & 0 & 5 \\ 0 & 2 & 1 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 2 & 0 & 0 & 1 & 0 & 5 \end{bmatrix},$$

and

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1/2 \end{bmatrix} \cdot \begin{bmatrix} 1 & -4 & -3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 3 & 1 & 0 & 0 & 0 & 6 \\ 0 & -3 & 2 & 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 2 & 0 & 0 & 1 & 0 & 5 \\ 0 & 2 & 1 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1/2 & 0 & 0 & 0 & 1/2 & 2 \end{bmatrix}.$$

In fact, if we combine all five sets of multipliers into a single matrix, i.e., if we let

$$\mathbf{P}_1 \equiv \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 3/2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1/2 \end{bmatrix},$$

then all of the above operations can be consolidated into a single matrix multiplication, namely

$$\begin{aligned} \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 3/2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1/2 \end{bmatrix} & \cdot \begin{bmatrix} 1 & -4 & -3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 3 & 1 & 0 & 0 & 0 & 6 \\ 0 & -3 & 2 & 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 2 & 0 & 0 & 1 & 0 & 5 \\ 0 & 2 & 1 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \\ & = \begin{bmatrix} 1 & 0 & -1 & 0 & 0 & 0 & 2 & 8 \\ 0 & 0 & 2 & 1 & 0 & 0 & -1 & 2 \\ 0 & 0 & 7/2 & 0 & 1 & 0 & 3/2 & 9 \\ 0 & 0 & 2 & 0 & 0 & 1 & 0 & 5 \\ 0 & 1 & 1/2 & 0 & 0 & 0 & 1/2 & 2 \end{bmatrix}; \end{aligned}$$

and the outcome of this multiplication is precisely the tableau produced by the first pivot in the Simplex algorithm.

Now, if we denote the matrix at the right-hand side of the above display as $\mathbf{T}_1$, then what we have is that

$$\mathbf{P}_1 \cdot \mathbf{T}_I = \mathbf{T}_1.$$

In words, this means that the Simplex tableau $\mathbf{T}_1$, obtained after the first pivot, is simply the product of the matrix $\mathbf{P}_1$ and the initial tableau $\mathbf{T}_I$.

Continuing in this manner, and using similar notation, we see that the combined effects of k consecutive pivots, starting with the initial tableau $\mathbf{T}_I$, can be conceptualized as

$$\mathbf{P}_k \cdot \cdots \cdot \mathbf{P}_1 \cdot \mathbf{T}_I = \mathbf{T}_k.$$

In particular, if we are interested in the final tableau, which we denote by $\mathbf{T}_F$, then we have

$$\mathbf{P} \cdot \mathbf{T}_I = \mathbf{T}_F,$$

where $\mathbf{P}$ is defined as the product of the string of matrices (of the form $\mathbf{P}_k \cdot \cdots \cdot \mathbf{P}_1$) that correspond to individual pivots that, together, lead to the final tableau. In other words, we have arrived at the conclusion that there exists a matrix $\mathbf{P}$ such that $\mathbf{P} \cdot \mathbf{T}_I = \mathbf{T}_F$; and that the matrix $\mathbf{P}$ fully captures the combined effects of all successive pivots. This important observation will be refered to as **fundamental insight**.

Observe, however, that the fundamental insight is essentially useless unless the matrix $\mathbf{P}$ is somehow available to us. So, the next question is: How does one determine the entries in $\mathbf{P}$? A little bit of reflection now leads us to the (unfortunate) realization that the Simplex algorithm itself can, in fact, be viewed as a procedure for generating the matrix $\mathbf{P}$. In other words, there is no free lunch. However, we will show that the story becomes quite different if we had gone through the solution of a linear program once.

Let us again return to the linear program above. Since we did solve that problem to optimality in an earlier section, the final tableau for this problem is available to us. This tableau is reproduced below.

z	x_1	x_2	s_1	s_2	s_3	s_4	
1	0	0	1/2	0	0	3/2	9
0	0	1	1/2	0	0	−1/2	1
0	0	0	−7/4	1	0	13/4	11/2
0	0	0	−1	0	1	1	3
0	1	0	−1/4	0	0	3/4	3/2

In our current notation, this means that we have

$$\mathbf{T}_F = \begin{bmatrix} 1 & 0 & 0 & 1/2 & 0 & 0 & 3/2 & 9 \\ 0 & 0 & 1 & 1/2 & 0 & 0 & -1/2 & 1 \\ 0 & 0 & 0 & -7/4 & 1 & 0 & 13/4 & 11/2 \\ 0 & 0 & 0 & -1 & 0 & 1 & 1 & 3 \\ 0 & 1 & 0 & -1/4 & 0 & 0 & 3/4 & 3/2 \end{bmatrix}.$$

Therefore, in the relation $\mathbf{P} \cdot \mathbf{T}_I = \mathbf{T}_F$, both $\mathbf{T}_I$ and $\mathbf{T}_F$ are explicitly known to us. This naturally suggests that we might be able to identify $\mathbf{P}$ more easily. It turns out that this indeed is possible. We will next describe two additional properties of matrix multiplications that will help us identify $\mathbf{P}$.

An $n \times n$ square matrix is called an identity matrix if all of its diagonal entries are equal to 1 and all of its off-diagonal entries are equal to 0. Such a matrix will be denoted by $\mathbf{I}_n$. For example, a $3 \cdot 3$ identity matrix assumes the form:

$$\mathbf{I}_3 \equiv \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

It is easily seen that if we are given a matrix $\mathbf{A}$ with n columns, then $\mathbf{A} \cdot \mathbf{I}_n = \mathbf{A}$. That is, (post) multiplying a given matrix by $\mathbf{I}_n$ will not change the identity of that matrix. (This is why we call $\mathbf{I}_n$ the identity matrix.)

Next, we will revisit the matrices $\mathbf{A}_1$ and $\mathbf{A}_2$, and use them to illustrate a slightly different way to describe a matrix multiplication. First, we will view the matrix $\mathbf{A}_2$ as a collection of three 3×1 matrices, or columns. These columns are explicitly given by

$$\mathbf{c}_1 \equiv \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}, \ \mathbf{c}_2 \equiv \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}, \text{ and } \mathbf{c}_3 \equiv \begin{bmatrix} 4 \\ 2 \\ -2 \end{bmatrix}.$$

Next, recall that the first column in $\mathbf{A}_3$ (the product of $\mathbf{A}_1$ and $\mathbf{A}_2$) is

$$\begin{bmatrix} 3 \\ 3 \end{bmatrix}.$$

It is easily seen that this column can be computed via $\mathbf{A}_1 \cdot \mathbf{c}_1$, i.e.,

$$\begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix};$$

and similarly, that the remaining two columns in $\mathbf{A}_3$ can be computed via $\mathbf{A}_1 \cdot \mathbf{c}_2$ and $\mathbf{A}_1 \cdot \mathbf{c}_3$. More formally, we have

$$\begin{aligned} \mathbf{A}_3 &\equiv \mathbf{A}_1 \cdot \mathbf{A}_2 \\ &= \mathbf{A}_1 \cdot \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \mathbf{c}_3 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{A}_1 \cdot \mathbf{c}_1 & \mathbf{A}_1 \cdot \mathbf{c}_2 & \mathbf{A}_1 \cdot \mathbf{c}_3 \end{bmatrix}; \end{aligned}$$

that is, the matrix $\mathbf{A}_3$ can be generated one column at a time, by executing a sequence of matrix products of the form $\mathbf{A}_1 \cdot \mathbf{c}_j$ with $j = 1, 2, 3$.

We now return to our example. Observe that in the initial tableau, the columns associated with the variables z, s_1, s_2, s_3, and s_4 constitute a $5 \cdot 5$ identity matrix, $\mathbf{I}_5$. Therefore, if we focus our attention on these five columns only, then, as a consequence of $\mathbf{P} \cdot \mathbf{T}_I = \mathbf{T}_F$ (the fundamental insight) and the alternative description of matrix multiplication above, the corresponding five columns in the final tableau must equal to $\mathbf{P} \cdot \mathbf{I}_5$. Since $\mathbf{P} \cdot \mathbf{I}_5 = \mathbf{P}$, we can now identify $\mathbf{P}$ as the matrix defined by these five columns in the final tableau. More explicitly, this simply means that since

$$\begin{aligned} \mathbf{P} \quad \cdot & \begin{bmatrix} 1 & - & - & 0 & 0 & 0 & 0 & - \\ 0 & - & - & 1 & 0 & 0 & 0 & - \\ 0 & - & - & 0 & 1 & 0 & 0 & - \\ 0 & - & - & 0 & 0 & 1 & 0 & - \\ 0 & - & - & 0 & 0 & 0 & 1 & - \end{bmatrix} \\ = & \begin{bmatrix} 1 & - & - & 1/2 & 0 & 0 & 3/2 & - \\ 0 & - & - & 1/2 & 0 & 0 & -1/2 & - \\ 0 & - & - & -7/4 & 1 & 0 & 13/4 & - \\ 0 & - & - & -1 & 0 & 1 & 1 & - \\ 0 & - & - & -1/4 & 0 & 0 & 3/4 & - \end{bmatrix}, \end{aligned}$$

where the $-$'s represent ignored entries, we must have

$$\mathbf{P} = \begin{bmatrix} 1 & 1/2 & 0 & 0 & 3/2 \\ 0 & 1/2 & 0 & 0 & -1/2 \\ 0 & -7/4 & 1 & 0 & 13/4 \\ 0 & -1 & 0 & 1 & 1 \\ 0 & -1/4 & 0 & 0 & 3/4 \end{bmatrix}.$$

In conclusion, we have shown that an important consequence of the fundamental insight is that if a linear program has been solved to optimality once, then the matrix $\mathbf{P}$ can be read out directly from the final tableau. Applications of this result to sensitivity analysis will be discussed in the next section.

2 Sensitivity Analysis: An Example

Consider the linear program:

$$
\begin{array}{llrrrrcll}
\text{Maximize} & z = & -5x_1 & +5x_2 & +13x_3 & & & & \\
\text{Subject to:} & & & & & & & & \\
 & & -x_1 & +x_2 & +3x_3 & \leq & 20 & & (1) \\
 & & 12x_1 & +4x_2 & +10x_3 & \leq & 90 & & (2)
\end{array}
$$

$$x_1,\, x_2,\, x_3 \geq 0\,.$$

After introducing two slack variables s_1 and s_2 and executing the Simplex algorithm to optimality, we obtain the following final set of equations:

$$
\begin{array}{lrrrrrcrl}
z & & & +2x_3 & +5s_1 & & = & 100\,, & (0) \\
 & -x_1 & +x_2 & +3x_3 & +s_1 & & = & 20\,, & (1) \\
 & 16x_1 & & -2x_3 & -4s_1 & +s_2 & = & 10\,. & (2)
\end{array}
$$

Our task is to conduct sensitivity analysis by *independently* investigating each of a set of nine changes (detailed below) in the original problem. For each change, we will use the fundamental insight to revise the final set of equations (in tableau form) to identify a new solution and to test the new solution for feasibility and (if applicable) optimality.

We will first recast the above equation systems into the following pair of initial and final tableaus.

Initial Tableau:

Basic Variable	z	x_1	x_2	x_3	s_1	s_2	
	1	5	−5	−13	0	0	0
s_1	0	−1	1	3	1	0	20
s_2	0	12	4	10	0	1	90

Final Tableau:

Basic Variable	z	x_1	x_2	x_3	s_1	s_2	
	1	0	0	2	5	0	100
x_2	0	−1	1	3	1	0	20
s_2	0	16	0	−2	−4	1	10

The basic variables associated with this final tableau are x_2 and s_2; therefore, the current basic feasible solution is $(x_1,\, x_2,\, x_3,\, s_1,\, s_2) = (0,\, 20,\, 0,\, 0,\, 10)$, which has an objective function value of 100.

An inspection of the initial tableau shows that the columns associated with z, s_1, and s_2 form a $3 \cdot 3$ identity matrix. Therefore, the $\mathbf{P}$ matrix will come from the corresponding columns in the final tableau. That is, we have

$$
\mathbf{P} = \begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix};
$$

and the final tableau equals the matrix product of this $\mathbf{P}$ and the initial tableau, i.e., $\mathbf{T}_F = \mathbf{P} \cdot \mathbf{T}_I$.

Our basic approach for dealing with parameter changes in the original problem is in two steps. In the first step, we will revise the final tableau by multiplying the *same* $\mathbf{P}$ to the new initial tableau; in other words, despite a revision in $\mathbf{T}_I$, we intend to follow the original sequence of pivots. After producing a revised $\mathbf{T}_F$, we will, in the second step, take the revised $\mathbf{T}_F$ as the starting point and initiate any necessary further analysis of the revised problem. We now begin a detailed sensitivity analysis of this problem:

a) Change the right-hand side of constraint (1) to 30. Denote the right-hand-side constants in the original constraints as b_1 and b_2. Then, the proposed change is to revise b_1 from 20 to 30, while retaining the original value of b_2 at 90. With this change, the RHS column in the initial tableau becomes

$$\begin{bmatrix} 0 \\ 30 \\ 90 \end{bmatrix}.$$

Since the rest of the columns in the initial tableau stays the same, the only necessary revision in $\mathbf{T}_F$ will be in the RHS column. To determine this new RHS column, we multiply $\mathbf{P}$ to the above new column to obtain:

$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 30 \\ 90 \end{bmatrix} = \begin{bmatrix} 150 \\ 30 \\ -30 \end{bmatrix}.$$

Since the basic variables in the final tableau are x_2 and s_2, the solution associated with the revised $\mathbf{T}_F$ is $(x_1, x_2, x_3, s_1, s_2) = (0, 30, 0, 0, -30)$. With a negative value for s_2, this (basic) solution is not feasible.

Geometrically speaking, increasing the value of b_1 from 20 to 30 means that we are relaxing the first inequality constraint. Relaxing a constraint is tantamount to enlarging the feasible set; therefore, one would expect an improved optimal objective function value. The fact that the revised solution above is not feasible is not a contradiction to this statement. It only means that additional work is necessary to determine the new optimal solution.

What causes the infeasibility of the new solution? Recall that the original optimal solution is $(x_1, x_2, x_3, s_1, s_2) = (0, 20, 0, 0, 10)$. Since x_1, x_3, and s_1 are serving as nonbasic variables, the defining equations for this solution are: $x_1 = 0$, $x_3 = 0$, and $-x_1 + x_2 + 3x_3 = 20$. Now, imagine an attempt to increase the RHS constant of the last equation from 20 to $20+\delta$ (say) while maintaining these three equalities. As we increase δ (from 0), we will trace out a family of solutions which exits the feasible region with large δ.

More formally, suppose the original RHS column is revised to

$$\begin{bmatrix} 0 \\ 20+\delta \\ 90 \end{bmatrix}; \text{ or alternatively, to } \begin{bmatrix} 0 \\ 20 \\ 90 \end{bmatrix} + \begin{bmatrix} 0 \\ \delta \\ 0 \end{bmatrix}.$$

Then, after premultiplying this new column by $\mathbf{P}$, we obtain

$$\begin{aligned}
\begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 20+\delta \\ 90 \end{bmatrix}
&= \begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \cdot \left(\begin{bmatrix} 0 \\ 20 \\ 90 \end{bmatrix} + \begin{bmatrix} 0 \\ \delta \\ 0 \end{bmatrix} \right) \\
&= \begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 20 \\ 90 \end{bmatrix} + \begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ \delta \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} 100 \\ 20 \\ 10 \end{bmatrix} + \begin{bmatrix} 5\delta \\ \delta \\ -4\delta \end{bmatrix} \\
&= \begin{bmatrix} 100+5\delta \\ 20+\delta \\ 10-4\delta \end{bmatrix}.
\end{aligned}$$

Hence, with $\delta = 10$, we indeed have $s_2 = -30$, which means that the original inequality constraint $12x_1 + 4x_2 + 10x_3 \leq 90$ is violated. Moreover, this calculation also shows that in order for $10 - 4\delta$ to remain nonnegative, δ cannot exceed $5/2$. In other words, at $\delta = 5/2$, the family of solutions $(0,\ 20+\delta,\ 0,\ 0,\ 10-4\delta)$ "hits" the constraint equation $12x_1 + 4x_2 + 10x_3 = 90$; and therefore, progressing further will produce solutions that are outside the feasible set.

Interestingly, our analysis above holds even if we allow δ to assume a negative value. Such a case corresponds to a tightening of the constraint $-x_1 + x_2 + 3x_3 \leq 20$. A quick inspection of

$$\begin{bmatrix} 100 + 5\delta \\ 20 + \delta \\ 10 - 4\delta \end{bmatrix}$$

shows that x_2 is reduced to 0 when δ reaches -20. It follows that in order to maintain feasibility, and hence optimality (since the optimality test is not affected by a change in the RHS column), of solutions of the form $(0,\ 20 + \delta,\ 0,\ 0,\ 10 - 4\delta)$, the value of δ must stay within the range $[-20,\ 5/2]$.

Another important observation regarding the above calculation is that the *optimal* objective-function value will increase from 100 to $100 + 5\delta$, provided that δ is sufficiently small (so that we remain within the feasible set). If we interpret the value of b_1 as the availability of a resource, then this observation implies that for every additional unit of this resource, the optimal objective-function value will increase by 5. Thus, from an economics viewpoint, we will be unwilling to pay more than 5 (dollars) for an additional unit of this resource. For this reason, the value 5 is called the **shadow price** of this resource. It is interesting to note that the shadow price of the first resource (5, in this case) can be read directly from the top entry in the second column of $\mathbf{P}$.

b) Change the right-hand side of constraint (2) to 70: Since the original value of b_2 is 90, this is an attempt to reduce the availability of the second resource by 20. The analysis is similar to that in part (a). Again, we will write the new RHS column in the initial tableau as

$$\begin{bmatrix} 0 \\ 20 \\ 90 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \delta \end{bmatrix},$$

where δ is targeted to assume the value -20. After premultiplying this new column by $\mathbf{P}$, we obtain

$$\begin{aligned} \begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \cdot & \left(\begin{bmatrix} 0 \\ 20 \\ 90 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \delta \end{bmatrix} \right) \\ &= \begin{bmatrix} 100 \\ 20 \\ 10 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \delta \end{bmatrix} \\ &= \begin{bmatrix} 100 \\ 20 \\ 10 + \delta \end{bmatrix}. \end{aligned}$$

Hence, for all δ within the range $[-10,\ \infty)$, solutions of the form $(0,\ 20,\ 0,\ 0,\ 10 + \delta)$ will remain optimal.

With the particular choice of $\delta = -20$, we have

$$\begin{bmatrix} 100 \\ 20 \\ 10 + \delta \end{bmatrix} = \begin{bmatrix} 100 \\ 20 \\ -10 \end{bmatrix}.$$

It follows that the new solution (0, 20, 0, 0, -10) is infeasible. As in part (a), we will not attempt to derive a new optimal solution.

The shadow price of the second resource can be read directly from the top entry in the third column of $\mathbf{P}$. In this case, it is given by 0. That the shadow price of the second resource is equal to 0 is expected. It is a consequence of the fact that in the current optimal solution, we have $s_2 = 10$ and hence there is already an excess in the supply of the second resource. In fact, we will have an over supply as long as the availability of the second resource is no less than 80 (which corresponds to $\delta = -10$).

c) Change b_1 and b_2 to 10 and 100, respectively. Again, we will first consider a revision of the RHS column in $\mathbf{T}_I$ of the form:

$$\begin{bmatrix} 0 \\ 20 \\ 90 \end{bmatrix} + \begin{bmatrix} 0 \\ \delta_1 \\ \delta_2 \end{bmatrix},$$

where δ_1 and δ_2 are two independent changes. After premultiplying this new column by $\mathbf{P}$, we obtain

$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \cdot \left(\begin{bmatrix} 0 \\ 20 \\ 90 \end{bmatrix} + \begin{bmatrix} 0 \\ \delta_1 \\ \delta_2 \end{bmatrix} \right) = \begin{bmatrix} 100 + 5\delta_1 \\ 20 + \delta_1 \\ 10 - 4\delta_1 + \delta_2 \end{bmatrix}.$$

With $\delta_1 = -10$ and $\delta_2 = 10$, the new RHS column in $\mathbf{T}_F$ is:

$$\begin{bmatrix} 50 \\ 10 \\ 60 \end{bmatrix}.$$

Since the new solution $(x_1, x_2, x_3, s_1, s_2) = (0, 10, 0, 0, 60)$ is feasible, it is also optimal. The new optimal objective-function value is 50.

d) Change the coefficient of x_3 (a nonbasic variable) in the objective function to $c_3 = 8$ (from $c_3 = 13$). Consider a revision in the value of c_3 by δ; that is, let $c_3 = 13 + \delta$. Then, the x_3-column in $\mathbf{T}_I$ is revised to

$$\begin{bmatrix} -13 - \delta \\ 3 \\ 10 \end{bmatrix}; \text{ or alternatively, to } \begin{bmatrix} -13 \\ 3 \\ 10 \end{bmatrix} + \begin{bmatrix} -\delta \\ 0 \\ 0 \end{bmatrix}.$$

From the fundamental insight, the corresponding revision in the x_3-column in $\mathbf{T}_F$ is

$$\begin{aligned} \begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \cdot \left(\begin{bmatrix} -13 \\ 3 \\ 10 \end{bmatrix} + \begin{bmatrix} -\delta \\ 0 \\ 0 \end{bmatrix} \right) &= \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix} + \begin{bmatrix} -\delta \\ 0 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 2 - \delta \\ 3 \\ -2 \end{bmatrix}. \end{aligned}$$

Therefore, if $\delta = -5$, which corresponds to $c_3 = 8$, then the new x_3-column in $\mathbf{T}_F$ is explicitly given by

$$\begin{bmatrix} 2 - \delta \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 - (-5) \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ -2 \end{bmatrix}.$$

Observe that the x_3-column is the only column in $\mathbf{T}_F$ that requires a revision, the variable x_3 is nonbasic, and the coefficient of x_3 in the revised R_0 is positive (7, that is). It follows that the original optimal solution $(x_1, x_2, x_3, s_1, s_2) = (0, 20, 0, 0, 10)$ remains optimal.

More generally, an inspection of the top entry in the new x_3-column,

$$\begin{bmatrix} 2-\delta \\ 3 \\ -2 \end{bmatrix},$$

reveals that the original optimal solution will remain optimal for all δ such that $2 - \delta \geq 0$, i.e., for all δ in the range $(-\infty, 2]$.

e) Change the coefficients in x_1 (a nonbasic variable) column. Change c_1 to -2, a_{11} to 0, and a_{21} to 5. This means that the x_1-column in $\mathbf{T}_I$ is revised from

$$\begin{bmatrix} 5 \\ -1 \\ 12 \end{bmatrix} \text{ to } \begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix}.$$

Since the corresponding new column in $\mathbf{T}_F$ is

$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix},$$

where the top entry, 2, is positive, and since x_1 is nonbasic in $\mathbf{T}_F$, we see that the original optimal solution remains optimal.

f) Change the coefficients in x_2 (a basic variable) column. Change c_2 to 6, a_{12} to 2, and a_{22} to 5. This means that the x_2-column in $\mathbf{T}_I$ is revised from

$$\begin{bmatrix} -5 \\ 1 \\ 4 \end{bmatrix} \text{ to } \begin{bmatrix} -6 \\ 2 \\ 5 \end{bmatrix}.$$

The fundamental insight implies that the corresponding new x_2-column in $\mathbf{T}_F$ is

$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \cdot \begin{bmatrix} -6 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ -3 \end{bmatrix}.$$

The fact that this new column is no longer of the form

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

indicates that x_2 cannot serve as a basic variable in R_1. It follows that a pivot in the x_2-column is needed to restore x_2 back to the status of a basic variable. More explicitly, the revised final tableau is

Basic Variable	z	x_1	x_2	x_3	s_1	s_2	
	1	0	4	2	5	0	100
—	0	-1	2	3	1	0	20
s_2	0	16	-3	-2	-4	1	10

and we will execute a pivot with the x_2-column as the pivot column and R_1 as the pivot row. After this pivot, we obtain

Basic Variable	z	x_1	x_2	x_3	s_1	s_2	
	1	2	0	−4	3	0	60
x_2	0	−1/2	1	3/2	1/2	0	10
s_2	0	29/2	0	5/2	−5/2	1	40

.

Since x_3 now has a negative coefficient in R_0, indicating that the new solution is not optimal, the Simplex algorithm should be restarted to derive a new optimal solution (if any).

g) Introduce a new variable x_4 with $c_4 = 10$, $a_{14} = 3$, and $a_{24} = 5$. This means that we need to introduce the new x_4-column

$$\begin{bmatrix} -10 \\ 3 \\ 5 \end{bmatrix}$$

into the initial tableau. (The precise location of this new column is not important.) The corresponding new column in the final tableau will be

$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \cdot \begin{bmatrix} -10 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ -7 \end{bmatrix}.$$

Since this column has a positive entry at the top and since x_4 is nonbasic, the current optimal solution remains optimal. In an application, this means that there is insufficient incentive to engage in the new "activity" x_4.

h) Introduce a new constraint $2x_1 + 3x_2 + 5x_3 \leq 50$. After adding a new slack variable s_3, this inequality constraint becomes $2x_1 + 3x_2 + 5x_3 + s_3 = 50$. Next, we incorporate this equation into the final tableau to obtain

Basic Variable	z	x_1	x_2	x_3	s_1	s_2	s_3	
	1	0	0	2	5	0	0	100
–	0	−1	1	3	1	0	0	20
s_2	0	16	0	−2	−4	1	0	10
s_3	0	2	3	5	0	0	1	50

.

Observe that x_2 participates in the new equation and, therefore, cannot serve as the basic variable for R_1. To rectify this situation, we will execute the row operation $(-3) \cdot R_1 + R_3$. This yields

Basic Variable	z	x_1	x_2	x_3	s_1	s_2	s_3	
	1	0	0	2	5	0	0	100
x_2	0	−1	1	3	1	0	0	20
s_2	0	16	0	−2	−4	1	0	10
s_3	0	5	0	−4	−3	0	1	−10

.

With $s_3 = -10$, the new basic solution is not feasible.

i) Change constraint (2) to $10x_1 + 5x_2 + 10x_3 \leq 100$. With this revision, the initial tableau becomes

Basic Variable	z	x_1	x_2	x_3	s_1	s_2	
	1	5	−5	−13	0	0	0
s_1	0	−1	1	3	1	0	20
s_2	0	10	5	10	0	1	100

.

After premultiplying this by $\mathbf{P}$, we obtain the revised final tableau below.

Basic Variable	z	x_1	x_2	x_3	s_1	s_2	
	1	10	0	2	5	0	100
–	0	1	1	3	1	0	20
s_2	0	6	1	-2	-4	1	10

Observe that x_2 participates in R_2 and, therefore, cannot serve as the basic variable for R_1. To rectify this situation, we will execute the row operation $(-1) \cdot R_1 + R_2$. This yields

Basic Variable	z	x_1	x_2	x_3	s_1	s_2	
	1	10	0	2	5	0	100
x_2	0	1	1	3	1	0	20
s_2	0	5	0	-5	-5	1	-10

With $s_2 = -10$, the new basic solution is not feasible.

In conclusion, when RHS constants (cases a, b and c) or a nonbasic variable column (cases d and e) are modified we can make comments regarding sensitivity without additional iterations. When a basic variable is changed (case f) or a constraint / variable is added (cases g, h and i), in general we may need to do extra iterations.

3 Sensitivity Analysis with LINDO

Consider the linear program:

$$\begin{array}{lrcrrrcrl}
\text{Minimize} & z & = & 84x_1 & +72x_2 & +60x_3 & & & \\
\text{Subject to:} & & & & & & & & \\
 & & & 90x_1 & +20x_2 & +40x_3 & \geq & 200 & (1) \\
 & & & 30x_1 & +80x_2 & +60x_3 & \geq & 180 & (2) \\
 & & & 10x_1 & +20x_2 & +60x_3 & \geq & 150 & (3)
\end{array}$$

$$x_1,\, x_2,\, x_3 \geq 0\,.$$

This problem comes from Exercise 3.6-4. Briefly, the scenario that gives rise to this formulation is as follows. A farmer wishes to determine the quantities of the available types of feed, which are corn, tankage, and alfalfa, that should be given to each pig. Since pigs will eat any mix of these feed types, the objective is to determine which mix will meet a given set of basic nutritional requirements at a minimum cost. The decision variables are:

x_1 = kilograms of corn,
x_2 = kilograms of tankage, and
x_3 = kilograms of alfalfa.

The basic nutritional ingredients are: carbohydrates, protein, and vitamins; and their respective requirements are expressed in constraints (1)–(3) above. Finally, the costs per kilogram of feed types, 84, 72, and 60, are in cents.

Now, to launch LINDO and enter the problem as:

```
MIN 84 X1 + 72 X2 + 60 X3
ST
90 X1 + 20 X2 + 40 X3 > 200
30 X1 + 80 X2 + 60 X3 > 180
10 X1 + 20 X2 + 60 X3 > 150
END
```

where X1, X2, and X3 correspond, respectively, to x_1, x_2, and x_3. Next, click on the "Solve" menu and then select "Solve". Click on "Yes" in the "Do Range (Sensitivity) Analysis?" dialog box. The following report should now be displayed in the "Reports Window":

```
LP OPTIMUM FOUND AT STEP      2

        OBJECTIVE FUNCTION VALUE

        1)      241.7143

  VARIABLE        VALUE          REDUCED COST
        X1         1.142857          0.000000
        X2         0.000000         17.714285
        X3         2.428571          0.000000


       ROW   SLACK OR SURPLUS     DUAL PRICES
        2)         0.000000         -0.771429
        3)         0.000000         -0.485714
        4)         7.142857          0.000000

 NO. ITERATIONS=       2


 RANGES IN WHICH THE BASIS IS UNCHANGED:

                          OBJ COEFFICIENT RANGES
 VARIABLE         CURRENT        ALLOWABLE        ALLOWABLE
                   COEF          INCREASE         DECREASE
       X1       84.000000        51.000000        37.199997
       X2       72.000000         INFINITY        17.714285
       X3       60.000000        11.272726        22.666666

                          RIGHTHAND SIDE RANGES
      ROW         CURRENT        ALLOWABLE        ALLOWABLE
                    RHS          INCREASE         DECREASE
        2      200.000000        24.999998        80.000000
        3      180.000000       120.000000         6.000000
        4      150.000000         7.142857         INFINITY
```

We will interpret this output from top to bottom. The Simplex algorithm went through two iterations to produce the optimal solution. The optimal objective-function value is 241.7143 (cents). LINDO automatically introduces slack or surplus variables to convert inequality constraints into equalities. The numbering of the rows starts with the number 1, as opposed to 0. Thus, ROW 1 refers to the objective-function row, ROW 2 refers to (functional) constraint (1), and so on; correspondingly, the slack or surplus variables will also be numbered this way. For this problem, three surplus variables are introduced, and they are named SLK 2, SLK 3, and SLK 4 (LINDO uses SLK to denote both slack and surplus variables). In our standard notation, these correspond to s_1, s_2, and s_3; therefore, the optimal solution is $(x_1, x_2, x_3, s_1, s_2, s_3) = (1.142857, 0, 2.428571, 0, 0, 7.142857)$.

To explain the columns under REDUCED COST and DUAL PRICES, we need to refer to the final tableau. To generate the final tableau, click on the title bar at the top of the problem window (to return focus to that window), click on the "Reports" menu, and then click "Tableau". The final tableau now appears at the bottom of the previous "Reports Window". It is pasted below.

```
THE TABLEAU

     ROW  (BASIS)          X1         X2         X3  SLK     2  SLK     3
       1 ART            0.000     17.714      0.000      0.771      0.486
       2        X3      0.000      1.571      1.000      0.007     -0.021
       3        X1      1.000     -0.476      0.000     -0.014      0.010
       4 SLK     4      0.000     69.524      0.000      0.286     -1.190


     ROW   SLK     4
       1      0.000  -241.714
       2      0.000     2.429
       3      0.000     1.143
       4      1.000     7.143
```

This tableau is in our standard format, except that it overflows into a second line. Here, the variable ART refers to z. (Recall that z was artificially created for convenience.) The basic variables are X3 (for ROW 2), X1 (for ROW 3), and SLK 4 (for ROW 4); and they assume the corresponding values listed in the RHS column. Note, however, that the RHS constant in ROW 1 is listed as -241.714. The negative sign here is a consequence of converting a minimization problem into a maximization problem, which LINDO does automatically. (Working with maximization problems only simplifies interpretation.) Therefore, this negative sign should be reversed in the final report, which, again, the program does automatically. In ROW 1, the coefficients of X2, SLK 2, and SLK 3, the nonbasic variables, are all positive; this indicates that what we have is indeed the (unique) final tableau.

We now return to the report above. The values listed in the REDUCED COST column are taken from the coefficients of X1, X2, and X3 in ROW 1, in the final tableau. In other words, X1 and X3 have a reduced cost of 0, whereas X2 has a reduced cost of 17.714. Formally (for a maximization problem), the **reduced cost** for a nonbasic variable is defined as the amount by which the value of z will decrease if we increase the value of that nonbasic variable by 1 (while holding all other nonbasic variables at 0). The adjective "reduced" is used because such a cost is relative to a specific tableau, i.e., is from the viewpoint of the particular current basic feasible solution. The 1-unit increment in the nonbasic variable is nominal, in that we are only contemplating an increase, even when it is not feasible to do so. The reduced cost for a basic variable is defined as 0. Mechanically, this is because basic variables always have a coefficient of 0 in the objective-function row; and conceptually, this is because the basic variables are already "participating" in the current solution (and therefore we do not attempt to bring them into the basis).

For example, the final tableau tells us that it is not optimal to include any tankage in the mix. Moreover, if we insist on having tankage in the mix, then the cost per kilogram of tankage is 17.714 cents. Like the concept of shadow price, this cost is relative to our current optimal solution, i.e., it has nothing to do with the "market" cost of tankage (which is at 72 cents per kilogram).

Next, we move on to the values listed in the DUAL PRICES column. The term "dual prices" is equivalent to shadow prices. (Every linear program has an associated dual linear program, and the concept of dual prices originates from the dual linear program. We will not discuss the dual of a linear program, as it is a more-advanced topic.) Recall (from parts (a) and (b) in the previous example) that the shadow price associated with the RHS constant of an original constraint (or with the availability of a resource) is defined as the amount by which the optimal objective-function value will improve if we increase the value of that constant by 1. (Again, this 1-unit increment is nominal.) In the current problem, the original functional constraints are of the "$\geq$" type. For such a constraint, an increase in the RHS constant corresponds to a tightening of that constraint; hence, the increase will (typically) result in a degradation of the optimal objective-function value. Indeed, the reported dual prices for ROW 2 and ROW 3 are negative.

For example, for every unit of increase in the nutritional requirement for carbohydrates, the cost of the *optimal* mix will increase by 0.771429 cents. Similarly, the corresponding increase in cost associated with protein is 0.485714. That the dual price for vitamins equals 0 is a consequence of the fact that the optimal mix already exceeds the vitamins requirement by a margin of 7.142857 (that is, the surplus variable SLK 4 equals 7.142857 in the optimal solution).

Next, we examine the RANGES IN WHICH THE BASIS IS UNCHANGED. Here, THE BASIS refers to the optimal basis; and two sets of ranges are displayed, one for the original objective-function coefficients and one for the original RHS constants.

Consider the range for X2 first. Recall (from part (d) in the previous example) that if the objective-function coefficient of a variable is revised to a new value, then the revision is directly reflected in the coefficient of that variable in ROW 1 of the final tableau. In the example here, this means that if the cost of tankage is revised by δ, to $72+\delta$, then the coefficient of X2 in ROW 1 of the final tableau will be revised to $17.714+\delta$. Moreover, since X2 is nonbasic, no further revision in the final tableau is necessary. It follows that δ has to be less than -17.714 to make the new coefficient negative, i.e., to make it desirable to include tankage in the mix. Indeed, LINDO reports that the objective-function coefficient of X2 should stay inside the interval $[72-17.714285,\ 72+\infty)$, or $[54.285715,\ \infty)$, in order for the current basis, and solution, to remain optimal.

Naturally, the maximum possible decrease of 17.714 is identical to the reduced cost of X2. That these two values are in agreement is expected, as they are derived from two (different-sounding but) equivalent viewpoints.

Next, suppose the cost of corn is revised by δ, from 84 to $84+\delta$. A similar argument shows that this will imply a corresponding revision of the coefficient of X1 in ROW 1 to $0+\delta$. Since X1 is basic, doing so will disqualify X1 as the basic variable for ROW 3. Hence, as in part (f) of the previous example, one needs to execute a pivot in the X1-column to eliminate the new entry δ (assuming it is nonzero). This pivot will result in a new set of coefficients in ROW 1; and therefore, we need to limit the scope of δ to ensure that these new coefficients remain nonnegative (i.e., to ensure that the current basis remains optimal). A little bit of calculations (which we leave out) now shows that δ should stay within the interval $[-37.19997,\ 51.000]$; and this corresponds to the LINDO statement for the range of the objective-function coefficient of X1.

The specified range for the objective-function coefficient of X3 is interpreted similarly: If the cost of alfalfa is revised from 60 to $60+\delta$, then δ should stay within the interval $[-22.666666,\ 11.272726]$ in order for the current tableau (and solution) to remain optimal.

Finally, we examine the ranges for the original RHS constants. For ROW 2, LINDO reports that if the RHS constant 200, which is the carbohydrates requirement, is revised to $200+\delta$, then the current basis will remain optimal for all δ inside the interval $[-80.000000,\ 24.999998]$. Similarly, the scopes of δ for ROW 3 and ROW 4, or for protein and vitamins, are reported as $[-6.000000,\ 120.000000]$ and $[-\infty,\ 7.142857]$, respectively. The supporting calculations for these statements are similar to what we did in parts (a) and (b) in the previous example.

It is important to realize, however, that although the optimal basis stays the same as long as δ is within these individual ranges, the specific *values* of the optimal basic variables do depend on δ. Again, the details are similar to parts (a) and (b) in the previous example. From a practical viewpoint, the reported ranges for both the objective-function coefficients and the RHS constants are rather wide. This *insensitivity of the optimal basis* with respect to changes in the input parameters is remarkable.

4 Exercises

1. Textbook H-L pp. 98-99, 3.4-19. Once you have the formulation, input it into Excel or Lindo and solve the problem. Report the solution and the objective value.

2. Go back to the previous problem. Obtain the sensitivity report. Refer to the sensitivity report while answering the following questions.
   a) How much does the optimal cost change if we nominally increase the requirement of at least 380 calories (a right-hand-side change)?
   b) How much does the cost change if we increase the requirement of 50 miligram vitamins to 60 grams (a right hand side change)?
   c) If we were to forget about constraints and force the optimal amount of the vitamin supplement to increase (a change in the value of a variable), what would the rate of change be in the cost?

3. Textbook H-L p. 100, 3.6-1. Do part a), input the formulation into Excel or Lindo and solve the problem. Report the solution and the objective value.

4. Once you have the formulation for the previous problem, input it into Lindo and solve the problem. Then go to "Reports" menu and display the final optimal tableau. What is the **P** matrix for this problem?

5. For a hypothetical problem, suppose that

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & -2 \end{bmatrix}.$$

   a) What is the relationship between Row 0 (denoted by R_0^F) of the final tableau and Rows of the initial tableau (denoted by R_0^I, R_1^I, R_2^I and R_3^I)? In other words, what are the question marks in the following equality: $R_0^F = ?R_0^I + ?R_1^I + ?R_2^I + ?R_3^I$. (Hint: Read the fundamental insight.)
   b) If the right hand sides were 0, 5, 10, 6, what is the value of the optimal objective value? What are the values of basic variables corresponding to each equation?
   c) What is the largest decrease and increase we can do on the rhs for the second equation (currently 10) before the current solution becomes infeasible?

6. Suppose I plan to make investments according to the solution of the investment problem (Textbook H-L pp. 96-97. 3.4-13) however, I owe \$100,000 to mafia due at the beginning of year 3. What is the maximum amount I can pay without changing the original optimal basis?